

CSI131 Discrete Structures 1
Tutorial Sequences and Summations
DURATION: 2 HOURS

Attempt all questions.

1. Is each sequence arithmetic? If so, find the common difference.
 - (a) $\{1, 2, 4, 8, 16, \dots\}$
 - (b) $\{-3, 1, 5, 9, 13, \dots\}$
2. Write the first five terms of the arithmetic sequence with $a_1 = 17$ and $d = -3$.
3. List the first four terms of the geometric sequence with $a_1 = 5$ and $r = -2$.
4. Is the sequence geometric? If so, find the common ratio.
 - (a) $\{1, 2, 4, 8, 16, \dots\}$
 - (b) $\{48, 12, 4, 2, \dots\}$
5. Write the first 5 terms of the sequence defined by the following explicit formula for $n \geq 1$:
 - (a) $-(-5)^{n-1}$
 - (b) $\frac{2^n}{n^3}$
 - (c) $a_n = \frac{(-1)^n n^2}{n+1}$
6. Evaluate the following series:
 - (a) $\sum_{r=1}^8 r(r+1)$
 - (b) $\sum_{k=3}^{10} (5k+2)$
 - (c) $\sum_{j=0}^8 (2^{j+1} - 2^j)$
 - (d) $\sum_{j=0}^8 (2.3^j + 3.2^j)$
7. Write the first six terms of the sequence defined by the recursive formula and initial conditions:

(a) $a_n = 3a_{n-1} + 4a_{n-2}, \quad a_1 = 1, a_2 = 2$

(b) $a_n = a_{n-1} + a_{n-2}, \quad a_0 = 2, a_1 = -1$

8. Write the first five terms of the sequence defined by the explicit formula

$$a_n = \frac{5n}{(n+2)!}.$$

9. Write an explicit formula for the n^{th} term of each sequence.

(a) $\{3, 6, 11, 18, 27, 38, 51, \dots\}$

(b) $\{-\frac{2}{11}, -\frac{3}{13}, -\frac{4}{15}, -\frac{5}{17}, -\frac{6}{19}, \dots\}$

(c) $\{-\frac{2}{25}, -\frac{2}{125}, -\frac{2}{625}, -\frac{2}{3125}, -\frac{2}{15625}, \dots\}$

(d) $\{e^4, e^5, e^6, e^7, e^8, \dots\}$

(e) $\{9, -81, 729, -6561, 59049, \dots\}$